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PROJECT REPORT

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Blockchain Based Farm Marketplace for Transparent Agricultural Trade

Submitted in partial fulfillment of the requirements for the degree

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE & ENGINEERING

Submitted by

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ST JOSEPH ENGINEERING COLLEGE

An Autonomous Institution

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Vamanjoor, Mangaluru – 575028, Karnataka

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DECLARATION

We hereby declare that the entire work embodied in this Project Report titled **"Blockchain Based Farm Marketplace for Transparent Agricultural Trade"** has been carried out by us at St Joseph Engineering College, Mangaluru under the supervision of **Ms Abhaya D K**, for the award of **Bachelor of Engineering** in **Computer Science & Engineering**. This report has not been submitted to this or any other University for the award of any other degree.

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Abstract

Agricultural supply chains have long suffered from a lack of transparency, fair pricing mechanisms, and reliable traceability. Farmers in developing countries, particularly smallholder farmers in India, face significant challenges in obtaining equitable prices for their produce due to the dominance of multiple intermediaries that create information asymmetry and reduce overall profitability.

This project proposes the development of a Blockchain Based Farm Marketplace for Transparent Agricultural Trade – a blockchain-based digital platform that directly connects farmers with buyers, eliminating unnecessary intermediaries. The system leverages blockchain technology to record all transactions in an immutable and tamper-proof ledger, and employs smart contracts to automate trade agreements between farmers and buyers without relying on traditional intermediaries.

The platform provides farmers with the ability to register, list agricultural products with pricing and quantity details, and receive payments securely. Buyers can browse listings, verify product origin through blockchain-based traceability records, and place orders directly. An admin interface enables platform monitoring and dispute resolution. The system is built as a cross-platform application using Expo (React Native) for the frontend, Node.js with Express.js and TypeScript for the backend, MongoDB for off-chain data, and Solidity smart contracts on Ethereum (Hardhat), with Ethers.js and a backend relayer service handling blockchain interaction.

The proposed system addresses the core challenges of intermediary dependency, information asymmetry, and lack of trust identified in the literature, and provides a practical, scalable, and user-friendly solution to improve farmer livelihoods and buyer confidence in agricultural trade.

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Introduction

1.1 Background

Blockchain technology has become one of the powerful new ideas in the world of digital tools. Blockchain technology gives us a way to keep records that cannot be changed and it lets many people share the information at the same time. In farming people need fast ways to trade. Farmers in places like India find it hard to get prices for their crops because many middlemen – agents, wholesalers and distributors – get involved. These middlemen create gaps in knowledge farmers' earnings and add extra delays to the trade process.

Traditional farm markets use central systems and paper records. These systems are easy to change can make mistakes. Are slow. Buyers and shoppers also find it hard to check where a product comes from and whether it is real. Because of that trust, in the food chain goes down. So we need a digital place that connects farmers and buyers straight shows clear price and trade details and lets people see where each product comes from. This new place should use an shared technology.

Blockchain technology can help us build this farm market. The new farm market will use the shared ledger to write down every trade that cannot be changed. Smart contracts made with Blockchain technology will run agreements automatically between farmers and buyers. The new farm market will also have a website and a phone app that work on devices so people can use it easily. By removing middlemen and showing clear and safe trades this new system will help farmers earn more help buyers trust what they buy and make the farm economy fairer.

1.2 Problem Statement

Agricultural supply chains traditionally involve multiple intermediaries such as wholesalers, distributors, and retailers positioned between farmers and buyers. These intermediaries often control pricing and market access, significantly reducing the profit received by farmers. In many cases, farmers lack direct access to market information, making it difficult for them to obtain fair prices for their products. Research on blockchain technology in supply chain management highlights that traditional agricultural markets suffer from limited transparency and inefficient transaction processes.

One of the big problem of traditional marketplace is that products cannot be traced.

Consumers and buyers cannot check the origin, quality or authenticity of goods. Without tracking it is hard to see how a product moves from farm to market and this can cause fraud, mislabeling and lower trust from customers. Traditional marketplaces also use a lot of paperwork and central systems that can be tampered with delayed or error-prone.

Therefore there is a need, for a marketplace that links farmers directly with buyers while keeping trade transparent, secure and traceable. The proposed Blockchain Based Farm Marketplace wants to solve these problems by using blockchain to record transactions cut out middlemen and give all participants reliable product data.

1.3 Objectives

The objectives of the proposed project are:

1. To design and develop a blockchain-based digital marketplace that directly connects farmers and buyers without intermediaries.
2. To implement smart contracts on a blockchain network to automate trade agreements and payment verification.
3. To provide blockchain-based product traceability that allows buyers to verify the origin and authenticity of agricultural products.
4. To build a secure, cross-platform mobile and web application accessible to users with minimal technical knowledge.
5. To record all platform transactions immutably on the blockchain, ensuring data integrity and transparency.

1.4 Scope

The scope of the Blockchain Based Farm Marketplace is to make agricultural trade clearer, safer and faster by using blockchain technology. The Blockchain Based Farm Marketplace wants to build a blockchain-based platform where farmers can meet buyers directly without middlemen.

The Blockchain Based Farm Marketplace lets farmers register and list products giving product details such as type, amount and price. Buyers can look at products check the details and buy them directly through the platform. Blockchain technology records every transaction in an clear ledger so that transaction data cannot be changed or tampered with.

The Blockchain Based Farm Marketplace also gives traceability of products by writing key information, about production, distribution and sales onto the blockchain. This lets buyers and consumers check the realness and source of products at any time.

The Blockchain Based Farm Marketplace is built as a digital trade help tool. Does not replace official financial or regulatory systems. Future upgrades might add IoT-based product monitoring. Link with government agricultural registries.

1.5 Definitions and Abbreviations

Blockchain: A decentralized digital ledger that records transactions across multiple computers in a secure and immutable manner. Once information is recorded on the blockchain, it cannot be modified or deleted, ensuring transparency and data integrity.

Smart Contract: A self-executing digital agreement stored on the blockchain that automatically executes transactions when predefined conditions are met, eliminating the need for intermediaries.

Agricultural Marketplace: A digital platform where farmers can list and sell their agricultural products and buyers can browse and purchase them directly.

Product Traceability: The ability to track the origin, movement, and history of agricultural products throughout the supply chain from farm to consumer.

Decentralized System: A system where data and control are distributed across multiple nodes rather than being managed by a single central authority, reducing the risk of manipulation or failure.

Cryptographic Security: The use of encryption techniques to protect data and ensure secure communication and transactions within a blockchain network.

DApp (Decentralized Application): An application that runs on a blockchain network rather than a centralized server, ensuring transparency and tamper resistance.

Gas Fee: A transaction cost charged on certain blockchain networks such as Ethereum to compensate for the computational resources used to process smart contracts.

Escrow: A mechanism in which the buyer's payment is held by the smart contract and released to the farmer only after delivery is confirmed.

Relayer Service: A backend service that signs and submits blockchain transactions on behalf of users, removing the need for users to manage wallets.

JWT (JSON Web Token): A compact token used to authenticate users and authorize access to protected resources.

Hardhat: An Ethereum development environment used to compile, deploy, test, and run local blockchain nodes for smart contracts.

Literature Survey

2.1 Blockchain Applications in Food Supply Chain Management – Ali et al. (2023)

The paper [1] presents a comprehensive review of blockchain applications in food supply chain management, examining how the technology can address long-standing challenges related to transparency, traceability, and trust. The main problem identified is that traditional food supply chains are fragmented, involving multiple intermediaries that reduce visibility and create opportunities for fraud, mislabeling, and inefficiency. The authors argue that these challenges are especially critical in agricultural and food systems, where product authenticity and safety directly affect consumer welfare.

The methodology involves a systematic literature review of existing studies on blockchain adoption in food and agricultural supply chains. The authors analyze key themes including the use of smart contracts for automating transactions, the role of blockchain in enhancing traceability from farm to consumer, and the challenges of integrating blockchain with existing supply chain technologies. The paper also categorizes blockchain platforms and their specific features relevant to food supply chain applications.

The results show that blockchain significantly improves traceability, reduces fraud, and enhances information sharing among supply chain participants. The study finds that smart contracts are particularly effective in automating payments and verifying product conditions, reducing dependency on intermediaries. The authors conclude that blockchain has strong potential to transform food supply chains by improving efficiency, transparency, and stakeholder trust.

However, several limitations are identified, including high implementation costs, the need for standardized data formats across the supply chain, and challenges related to scalability on public blockchain networks. The paper recommends future research focused on cost-effective blockchain architectures and integration with IoT and AI technologies for real-time supply chain monitoring.

2.2 Blockchain Adoption in Agri-Food Supply Chains – Nguyen et al. (2023)

The paper [2] conducts a systematic literature review of blockchain adoption in agri-food supply chains, focusing on identifying adoption drivers, barriers, and gaps in existing research. The central problem addressed is that despite growing interest in blockchain technology, its adoption in agriculture remains limited and inconsistent across different contexts. The authors highlight that farmers, processors, and retailers face different challenges that affect how blockchain is implemented and accepted within the agri-food sector.

The methodology is based on a structured review of peer-reviewed literature from major academic databases. The authors identify key themes including traceability, smart contract automation, data security, and stakeholder trust as central drivers of blockchain adoption. Barriers to adoption such as lack of technical infrastructure, limited digital literacy among farmers, regulatory uncertainty, and high transaction costs are also systematically analyzed.

The findings indicate that blockchain can improve transparency and traceability significantly but that adoption depends heavily on enabling factors such as government support, digital infrastructure, and user education. The paper identifies a significant research gap in studies focusing on smallholder farmers and developing country contexts, where the need for fair and transparent agricultural trade is greatest.

Limitations include the exclusion of grey literature and the rapid evolution of blockchain technology, which may render some reviewed studies outdated. The authors recommend future studies focusing on practical blockchain implementation frameworks suited to smallholder and developing country agricultural contexts, which aligns closely with the motivation of the proposed system.

2.3 Blockchain-Enabled Smart Agriculture – Patil and Bhosale (2024)

The paper [3] proposes a blockchain-enabled framework for smart agriculture that focuses on enhancing transparency and traceability in food systems. The main problem identified is that modern food systems suffer from a lack of reliable traceability mechanisms, making it difficult for consumers and regulators to verify the origin, quality, and handling of agricultural products. The authors argue that without trustworthy traceability, food fraud and mislabeling continue to undermine consumer confidence and market efficiency.

The methodology involves designing and evaluating a blockchain-based architecture that integrates smart contracts, IoT sensors, and decentralized data storage for end-to-end

traceability in agricultural production and distribution. The framework allows data to be captured at each stage of the supply chain – from planting and harvesting to processing and retail – and recorded immutably on the blockchain for verification by all participants.

The results demonstrate that the proposed framework significantly improves traceability accuracy and reduces the time required to identify the source of contamination or mislabeling in the supply chain. The system also improves consumer trust by providing accessible and verifiable product histories. The authors conclude that blockchain-enabled smart agriculture can contribute to both sustainability and cleaner production practices by reducing waste and improving accountability.

Limitations include high energy consumption associated with certain blockchain consensus mechanisms, the cost of IoT sensor deployment, and challenges in ensuring data accuracy at the point of entry. Future scope includes integrating AI-based analytics for predictive supply chain management and expanding the framework to cover cross-border agricultural trade.

2.4 Blockchain-Based Digital Agriculture Systems – Liu et al. (2024)

The paper [4] proposes a blockchain-based digital agriculture system designed to ensure secure and transparent supply chains through the integration of blockchain with Internet of Things (IoT) technologies. The main problem addressed is that existing digital agriculture systems lack sufficient security and transparency, making them vulnerable to data tampering, unauthorized access, and loss of trust among participants. The authors argue that combining blockchain's immutability with IoT's real-time data collection capabilities can create a more reliable and trustworthy agricultural data management system.

The methodology involves designing a layered architecture that connects IoT devices for data collection with a blockchain ledger for secure data storage and sharing. Smart contracts are used to automate data validation and transaction processes, ensuring that only verified and authorized data is recorded on the blockchain. The system is evaluated using test scenarios covering product registration, traceability queries, and transaction verification.

The results show that the proposed system achieves high data security, low latency in transaction processing, and reliable traceability across the agricultural supply chain. The integration of IoT with blockchain is shown to improve data accuracy and reduce the possibility of fraudulent data entry. The authors conclude that such systems are practical and viable for real-world deployment in precision agriculture and supply chain management.

Limitations include the high cost of IoT device deployment, energy consumption con-

cerns associated with blockchain consensus mechanisms, and dependency on network connectivity in rural agricultural environments. Future work is recommended to focus on lightweight blockchain protocols and energy-efficient consensus algorithms suitable for resource-constrained agricultural settings.

2.5 Blockchain and AI Integration for Agricultural Supply Chains – Singh et al. (2025)

The paper [5] proposes a next-generation agricultural supply chain framework that integrates blockchain technology with Artificial Intelligence to improve transparency, efficiency, and decision-making. The main problem identified is that while blockchain provides a secure and transparent ledger for supply chain data, it lacks the analytical intelligence to derive actionable insights from recorded data. Traditional supply chain systems are neither transparent nor adaptive, and existing blockchain-only solutions do not fully address the need for intelligent decision support in agricultural trade.

The methodology involves developing a hybrid architecture that combines blockchain for secure and immutable transaction recording with AI models for demand prediction, anomaly detection, and supply chain optimization. Smart contracts are used to automate routine processes, while AI algorithms analyze on-chain and off-chain data to provide recommendations for pricing, inventory management, and fraud detection. The framework is evaluated using a simulation environment with realistic agricultural supply chain datasets.

The results demonstrate that the integrated blockchain-AI system outperforms standalone blockchain and standalone AI solutions in terms of prediction accuracy, fraud detection rates, and overall supply chain efficiency. The authors conclude that the combination of blockchain and AI represents a significant advancement for agricultural supply chains, enabling both transparency and intelligence in a unified framework.

Limitations include the computational complexity of integrating AI with blockchain, challenges in ensuring data privacy when AI models are trained on blockchain-stored data, and the need for large annotated datasets for effective AI model training. Future directions include federated learning approaches for privacy-preserving AI in blockchain supply chains and real-time adaptive smart contracts that respond dynamically to AI predictions.

2.6 The Rise of Blockchain Technology in Agriculture and Food Supply Chains – Kamilaris et al. (2019)

The paper [6] surveys the emergence of blockchain technology in agriculture and food supply chains, examining both academic studies and industrial initiatives. The main problem addressed is that, while blockchain was attracting rapid interest, there was little consolidated understanding of where it could add value in agriculture, what projects already existed, and what obstacles stood in the way of wider adoption.

The methodology involves reviewing existing blockchain projects and research works related to agriculture, food traceability, and agricultural trade, and analyzing the benefits, challenges, and policy implications reported for each. The authors discuss how blockchain can improve food safety, transparency, and traceability, and how smart contracts and digital payments can simplify transactions between supply chain participants.

The results indicate that blockchain is a promising foundation for transparent and trustworthy agri-food systems, particularly for traceability and for reducing the role of intermediaries in transactions. However, the authors also highlight challenges including scalability, regulatory uncertainty, interoperability, the cost of adoption, and the limited technical skills of many agricultural stakeholders. These observations directly motivate the simple, wallet-free user experience targeted by the proposed system.

2.7 Blockchain-Based Traceability in Agri-Food Supply Chain Management – Caro et al. (2018)

The paper [7] presents AgriBlockIoT, a blockchain-based traceability solution for the agri-food supply chain that integrates Internet of Things devices with a decentralized ledger. The problem addressed is that traditional traceability systems depend on centralized databases controlled by a single party, which limits trust and transparency among the participants in the supply chain.

The methodology involves designing a fully decentralized architecture in which IoT sensor data is recorded on the blockchain and made available to all authorized participants. The authors implemented the solution on two different blockchain platforms, Ethereum and Hyperledger Sawtooth, and compared their behaviour to evaluate the practical feasibility of the approach.

The results show that a blockchain-based traceability system can be implemented in practice and that it offers tamper-resistant, auditable records of products from production

to consumption. Limitations include the dependence on IoT hardware, the limited scale of the evaluation, and the performance and cost differences between blockchain platforms. The proposed system takes a similar Ethereum-based approach but concentrates on the transactional layer without requiring dedicated sensor hardware.

2.8 Blockchain-Based Soybean Traceability in Agricultural Supply Chain – Salah et al. (2019)

The paper [8] proposes a blockchain-based solution for tracing soybeans through the agricultural supply chain using Ethereum smart contracts. The main problem addressed is the lack of trust, traceability, and accountability in traditional agricultural supply chains, where records are held by individual parties and can be altered or disputed.

The methodology involves using Ethereum smart contracts to manage and record all interactions and transactions among the participants in the supply chain, from the farmer through intermediaries to the final customer. The authors present the system design, the smart contract logic and its algorithms, and evaluate the solution in terms of cost, security, and design considerations.

The results demonstrate that smart contracts can automate interactions between participants and provide a transparent and tamper-resistant record of the product history without a trusted central authority. Limitations include the transaction cost of public Ethereum networks and the focus on a single crop. This work is closely related to the proposed system, which also uses Ethereum smart contracts, and extends the idea by adding an escrow-based payment flow and a complete cross-platform marketplace.

2.9 An Agri-Food Supply Chain Traceability System for China Based on RFID and Blockchain Technology – Tian (2016)

The paper [9] proposes an agri-food supply chain traceability system for China that combines Radio Frequency Identification (RFID) with blockchain technology. The problem addressed is that centralized traceability systems suffer from information asymmetry and a lack of trust between supply chain participants, which makes food safety incidents difficult to investigate.

The methodology combines Hazard Analysis and Critical Control Points (HACCP) principles, RFID-based data capture, and a blockchain-based decentralized ledger to build a system that records information at each stage of the agri-food supply chain. The decentralized design is intended to make the shared information transparent, traceable, and difficult to tamper with.

The results show that the combination of RFID and blockchain can improve information transparency and strengthen supply chain traceability and food safety management. Limitations include the cost of RFID infrastructure and the early-stage nature of the proposal. In contrast, the proposed marketplace avoids additional hardware and relies on blockchain transaction records linked to each listing.

2.10 Blockchain Technology for Agriculture: Applications and Rationale – Xiong et al. (2020)

The paper [10] reviews the applications of blockchain technology in agriculture and explains the rationale for its adoption. The problem addressed is that agricultural stakeholders face issues of trust, data transparency, and inefficient transactions, and there was a need to clarify where blockchain can genuinely help.

The methodology involves a review of blockchain applications in several agricultural areas, including agri-food supply chain traceability, agricultural insurance, and transactions in agricultural products. The authors discuss the technical features of blockchain, such as decentralization, immutability, and smart contracts, and relate them to the specific needs of each application area.

The findings suggest that blockchain can support more transparent and efficient agricultural trade and can reduce transaction costs by limiting the role of intermediaries. The authors also note challenges such as technical complexity, cost, and the need for supporting infrastructure and policy. These conclusions support the design choice of the proposed system to use smart contract based escrow for direct farmer-to-buyer transactions.

2.11 Comparison of Existing Systems

The following table compares the existing systems discussed in the literature survey based on their objectives, methodology, key findings, and limitations. It helps in understanding the strengths and gaps of previous works and in identifying the specific contributions of the proposed system.

Table 2.1: Comparison of Existing Systems

Author(s)	**Objective**	**Methodology**	**Key Findings**	**Limitations**
Ali et al. [1]	Review blockchain in food supply chains	Systematic review, smart contracts	Improves traceability and reduces fraud	High cost, scalability issues
Nguyen et al. [2]	Study blockchain adoption in agri-food	Structured review, adoption analysis	Adoption depends on infrastructure	Limited focus on smallholders
Patil & Bhosale [3]	Enhance traceability in food systems	Blockchain with IoT sensors	Improved traceability and consumer trust	High IoT deployment cost
Liu et al. [4]	Secure digital agriculture supply chains	Blockchain and IoT integration	High security and low latency	Rural connectivity dependency
Singh et al. [5]	Next-gen AI-blockchain supply chains	Blockchain with AI analytics	Improved prediction and fraud detection	Computational complexity
Kamilaris et al. [6]	Survey blockchain in agriculture and food	Review of projects and research works	Promising for transparency and traceability	Scalability, regulation, adoption cost
Caro et al. [7]	Decentralized agri-food traceability	Blockchain with IoT (Ethereum, Sawtooth)	Feasible tamper-resistant traceability	IoT hardware dependency
Salah et al. [8]	Trace soybeans in supply chain	Ethereum smart contracts	Transparent, automated interactions	Public network cost, single crop

Table 2.1: Comparison of Existing Systems (continued)

Author(s)	**Objective**	**Methodology**	**Key Findings**	**Limitations**
Tian [9]	Traceability system for China	RFID with blockchain, HACCP	Improved transparency and food safety	RFID infrastructure cost
Xiong et al. [10]	Review blockchain applications in agriculture	Application review and rationale	Supports efficient, transparent trade	Technical complexity, cost
Proposed System	Direct farmer-buyer marketplace	Blockchain, smart contracts, cross-platform app	Transparent trade, fair pricing, traceability	Internet dependency, relayer trust

2.12 Proposed System

Based on the analysis of existing literature it is clear that blockchain technology has been widely studied in areas like agricultural supply chain management, food safety traceability and smart agriculture. However there remains a gap when it comes to systems that connect farmers and buyers directly through a blockchain-based marketplace. Most current research focuses either on supply chain monitoring at the level or on traceability systems that rely heavily on IoT devices and expensive hardware. Few studies have looked into creating blockchain platforms that're easy to use and practical especially for smallholder farmers who want to sell their products directly to buyers without involving middlemen.

The proposed Blockchain Based Farm Marketplace for Transparent Agricultural Trade fills this gap. It introduces a marketplace built on blockchain technology where farmers can register list their agricultural products set their own prices and complete sales directly with buyers. The system uses contracts on a blockchain network to automate agreements and verify payments. This ensures that both farmers and buyers are protected, without the need for intermediaries like agents or wholesalers. Every transaction is recorded on the blockchain creating a unchangeable record. This provides an trustworthy system that builds confidence in the marketplace and strengthens accountability, between all parties involved.

The proposed system uses a hybrid architecture. It records trade data orders, escrow

payments and product history on the blockchain with smart contracts. It keeps user profiles and product details in an off-chain database called MongoDB. A backend relayer signs blockchain transactions for users so farmers and buyers can use the platform without handling cryptocurrency wallets.

Most current systems look mainly at traceability via sensors or at the supply chain level for big organizations. The proposed system however concentrates on the transaction side of trade. It gives blockchain advantages to farmers and buyers through a simple mobile and web app that works on many platforms. Traceability comes from blockchain transaction records that are tied to each product listing. Buyers can check origin and authenticity without hardware.

The proposed system moves agricultural blockchain technology forward by mixing transaction management, smart contract automation, product traceability and a direct farmer-to-buyer market into one platform. It tackles problems too many middlemen, uneven information and low transparency that were found in earlier studies. This practical scalable solution can truly help farmers earn more and give buyers trust, in agricultural trade.

Software Requirements Specification

3.1 Functional Requirements

3.1.1 User Registration

- The system shall allow farmers and buyers to create accounts by providing necessary details such as name, contact information, and login credentials.
- The system shall validate user information during the registration process.
- The system shall assign appropriate roles to users upon successful registration.

3.1.2 User Authentication

- The system shall provide a secure login mechanism for farmers, buyers, and administrators.
- The system shall authenticate users before allowing access to platform features.
- The system shall support secure session management and logout functionality.

3.1.3 Farmer Product Listing

- The system shall allow farmers to upload and manage details of their agricultural products including crop type, price, quantity, and description.
- The system shall allow farmers to update or remove their listed products at any time.
- The system shall display all active farmer product listings to buyers on the marketplace.

3.1.4 Product Browsing and Search

- The system shall allow buyers to browse available agricultural products listed by farmers.
- The system shall allow buyers to search and filter products by type, price, or availability.
- The system shall display product details including origin, price, and available quantity.

3.1.5 Blockchain Transaction Recording

- The system shall record all transactions between farmers and buyers on the blockchain ledger.
- The system shall ensure that recorded transaction data is immutable and tamper-proof.
- The system shall generate and provide unique transaction IDs for each completed transaction.

3.1.6 Smart Contract Execution

- The system shall use smart contracts to automate agreements between farmers and buyers.
- The system shall lock buyer payments in escrow at the time of purchase.
- Escrowed funds shall be released to the farmer only after the buyer confirms delivery.
- The system shall allow an order to be cancelled, with escrowed funds returned to the buyer.
- The system shall verify smart contract execution and provide confirmation to both parties upon completion.

3.1.7 Order Placement

- The system shall allow buyers to place orders for agricultural products through the platform.
- The system shall notify farmers when a new order is placed for their products.
- The system shall allow buyers to view and track the status of their orders.

3.1.8 Transaction History

- The system shall allow users to view their complete past transactions and order history.
- Transaction records shall be stored on the blockchain and remain accessible at any time.
- The system shall display transaction details including product name, quantity, price, and transaction date.

3.1.9 Product Traceability

- The system shall allow buyers to verify the origin and details of agricultural products through blockchain records.
- The system shall display traceability information linked to each product listing.
- Traceability data shall be stored on the blockchain and remain tamper-proof at all times.

3.1.10 Shopping Cart

- The system shall allow buyers to add multiple products to a cart.
- The system shall allow buyers to update or remove cart items and check out multiple items in a single flow.

3.1.11 Reviews and Ratings

- The system shall allow buyers to submit ratings and reviews for purchased products.

- The system shall display reviews and average ratings on product detail pages.

3.1.12 Government Schemes and AI Guidance

- The system shall provide farmers with information on government agricultural schemes.
- The system shall provide AI-based agricultural guidance in response to farmer queries (prototype).

3.1.13 Product Image Upload

- The system shall allow farmers to capture or select product images from the device camera or gallery.
- The system shall store uploaded images on a cloud media service and link them to product listings.

3.1.14 Admin Management

- The system shall allow administrators to manage users, products, and platform activities.
- Administrators shall be able to monitor transactions and resolve disputes between farmers and buyers.
- The system shall provide an admin dashboard displaying platform statistics and user activity.

3.2 Non-Functional Requirements

3.2.1 Security

- The system must ensure secure user authentication and protect transaction data using blockchain and encryption techniques.
- User passwords shall be encrypted and stored securely in the database.
- Access to transaction data and user information shall be restricted to authorized users only.
- The system shall use JWT-based authentication and role-based access control to restrict features by user role (farmer, buyer, administrator).

3.2.2 Reliability

- The platform shall operate consistently without failures during normal usage.
- All transaction records stored on the blockchain must remain accurate and accessible at all times.
- The system shall maintain complete data integrity during all operations.

3.2.3 Scalability

- The system shall support an increasing number of users, transactions, and product listings without performance degradation.
- The platform architecture shall allow future expansion of features and user capacity.

3.2.4 Usability

- The user interface shall be simple and intuitive for both farmers and buyers, including those with limited technical knowledge.
- Users shall be able to list products, browse listings, and complete transactions with minimal training.
- Users shall not be required to manage cryptocurrency wallets directly, as blockchain transactions are handled through a backend relayer service.

3.2.5 Availability

- The system shall be accessible to users at all times with minimal planned and unplanned downtime.
- The platform shall handle multiple concurrent users without service interruption.

3.2.6 Data Integrity

- All transaction data stored on the blockchain must remain accurate and tamper-proof.
- The system shall prevent unauthorized modification of product listings and transaction records.

3.2.7 Performance

- The system shall process blockchain transactions efficiently without long delays.
- Product information and transaction history shall be retrieved quickly and displayed to the user.

3.2.8 Maintainability

- The system architecture shall be modular to allow easy updates, bug fixes, and feature additions.
- Code shall be well-documented to support future development and maintenance.

3.3 User Interface Design

3.3.1 Farmer Interface

- **Login Page:** Allows farmers to securely authenticate and access the platform.
- **Product Listing Page:** Enables farmers to add, edit, and manage their agricultural product listings with details such as type, quantity, price, and description.
- **Transaction Dashboard:** Displays incoming order requests, order status, and completed transaction details.
- **Profile Management:** Allows farmers to update their personal information and account settings.
- **Image Capture/Upload:** Allows farmers to photograph or select product images while creating listings.
- **Government Schemes and AI Guidance:** Provides scheme information and agricultural guidance.

3.3.2 Buyer Interface

- **Login Page:** Allows buyers to access their account securely.
- **Product Marketplace:** Displays all available agricultural products listed by farmers with search and filter options.
- **Product Detail Page:** Shows complete product information including traceability data from the blockchain.
- **Order Placement Page:** Allows buyers to place orders and confirm purchases directly through smart contracts.
- **Order History Dashboard:** Displays all previous purchases, order status, and transaction details.
- **Shopping Cart:** Allows buyers to review selected items and check out multiple products together.
- **Reviews and Ratings:** Allows buyers to rate and review purchased products.

3.3.3 Admin Interface

- **Admin Dashboard:** Displays overall platform statistics, user activity, and transaction summaries.
- **User Management:** Allows administrators to view, approve, and manage farmer and buyer accounts.
- **Product Monitoring:** Enables administrators to monitor listed products, flag inappro-

priate listings, and oversee transactions.

3.4 Hardware and Software Requirements

3.4.1 Hardware Requirements

- **Processor:** Intel Core i5 or higher for efficient application processing.
- **RAM:** Minimum 8 GB RAM to support application execution and blockchain operations.
- **Storage:** At least 256 GB storage for system data and application files.
- **Internet Connection:** Stable broadband connection for accessing the platform and blockchain network.
- **Server System:** A dedicated or cloud-based server to handle backend operations and blockchain transactions.

3.4.2 Software Requirements

- **Operating System:** Windows 10 / Ubuntu Linux 20.04 or higher.
- **Programming Languages:** TypeScript for frontend and backend; Solidity for smart contract development.
- **Frontend Technologies:** Expo (React Native SDK 54) with Expo Router, React Native Web, and React Native Paper for cross-platform (Android, iOS, and Web) support.
- **Backend Framework:** Node.js and Express.js with TypeScript, JWT-based authentication, and bcrypt password hashing.
- **Blockchain Platform:** Ethereum with a local Hardhat node for development and testing.
- **Smart Contract Tools:** Hardhat for compilation, deployment, and testing.
- **Blockchain Interaction Library:** Ethers.js 6, used through a backend relayer service that signs transactions on behalf of users.
- **Database:** MongoDB with Mongoose for storing user profiles and product data off-chain.
- **Media Storage:** Cloudinary for product image uploads.
- **Development Environment:** Visual Studio Code as the primary code editor.

3.5 Performance Requirements

- The system shall process blockchain transactions and confirm smart contract execution within an acceptable timeframe.
- The platform shall support simultaneous access by multiple farmers and buyers without

significant performance degradation.

- Product listings and transaction history shall be loaded and displayed within a few seconds under normal network conditions.
- The system shall handle multiple concurrent product uploads and order placements without delays.

3.6 Design Constraints

- The system requires a stable internet connection to interact with the blockchain network; offline functionality is not supported.
- Transaction costs on public networks such as Ethereum mainnet may affect operational costs; the current implementation uses a local Hardhat network, where the relayer pays gas fees.
- Users require basic familiarity with digital platforms; blockchain wallets are managed by the backend relayer service.
- The backend relayer signs transactions on behalf of users, introducing a degree of trust and centralization dependency.
- Public blockchain networks may have inherent limitations in handling very high transaction volumes simultaneously.
- The system must remain compatible with the web technologies and blockchain frameworks selected during development.
- The platform is intended for agricultural trade support only and does not replace formal legal, financial, or regulatory systems.

3.7 Other Requirements

- The system shall maintain a complete and immutable history of all transactions recorded on the blockchain.
- Product listings and transaction records shall be presented in a clear, structured, and easily readable format.
- The system shall provide blockchain-based traceability information to help buyers verify product authenticity and origin.
- The platform shall comply with relevant data protection and privacy principles for the handling of personal user information.
- The system architecture shall support future extensions such as IoT-based product monitoring and advanced analytics dashboards.

System Design

4.1 Architecture Design

The proposed system follows a three-tier architecture consisting of the presentation layer, the application layer, and the blockchain layer. The presentation layer is a cross-platform application built with Expo (React Native), running on Android, iOS, and the web, and provides interfaces for farmers, buyers, and administrators. The application layer is built using Node.js, Express.js, and TypeScript, handling business logic, JWT-based authentication, role-based access control, and communication with the blockchain. The blockchain layer hosts Solidity smart contracts deployed on an Ethereum-compatible network (a local Hardhat node during development), ensuring immutable transaction recording, escrow-based payment handling, and automated agreement execution.

Ethers.js serves as the bridge between the backend and the blockchain. A backend relayer service signs and submits transactions on behalf of users, so farmers and buyers do not need to manage cryptocurrency wallets. Off-chain data such as user profiles, product metadata, carts, and reviews are stored in MongoDB, product images are stored on Cloudinary, and all payment-related transactions and traceability records are stored on the blockchain.

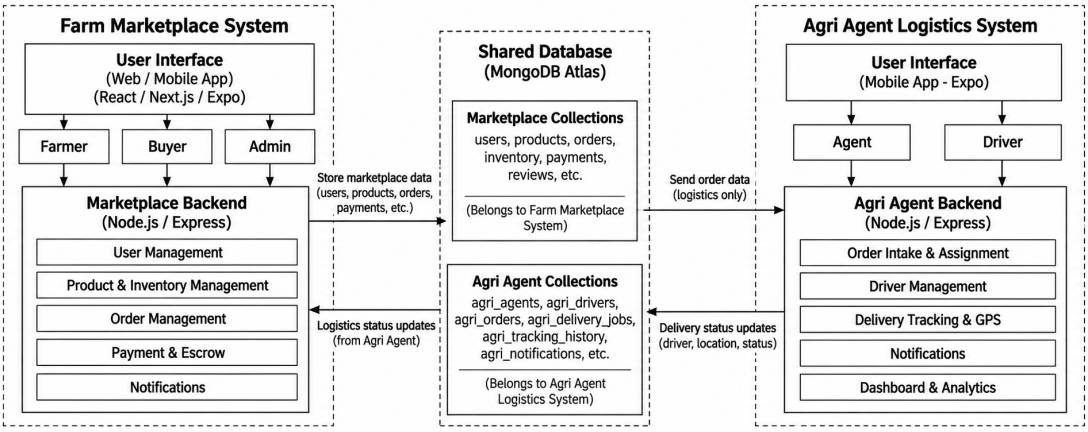


Figure 4.1: System Architecture Diagram

4.2 Use Case Diagram

The system involves three primary actors: Farmer, Buyer, and Administrator. The Farmer can register, log in, list products with images, update or remove listings, view orders, check transaction history, and access government scheme and AI guidance. The Buyer can register, log in, browse and search products, add items to a cart, view product traceability data, place

orders, confirm delivery, submit reviews, and access order history. The Administrator can manage user accounts, monitor listings, oversee transactions, and resolve disputes through the admin dashboard.

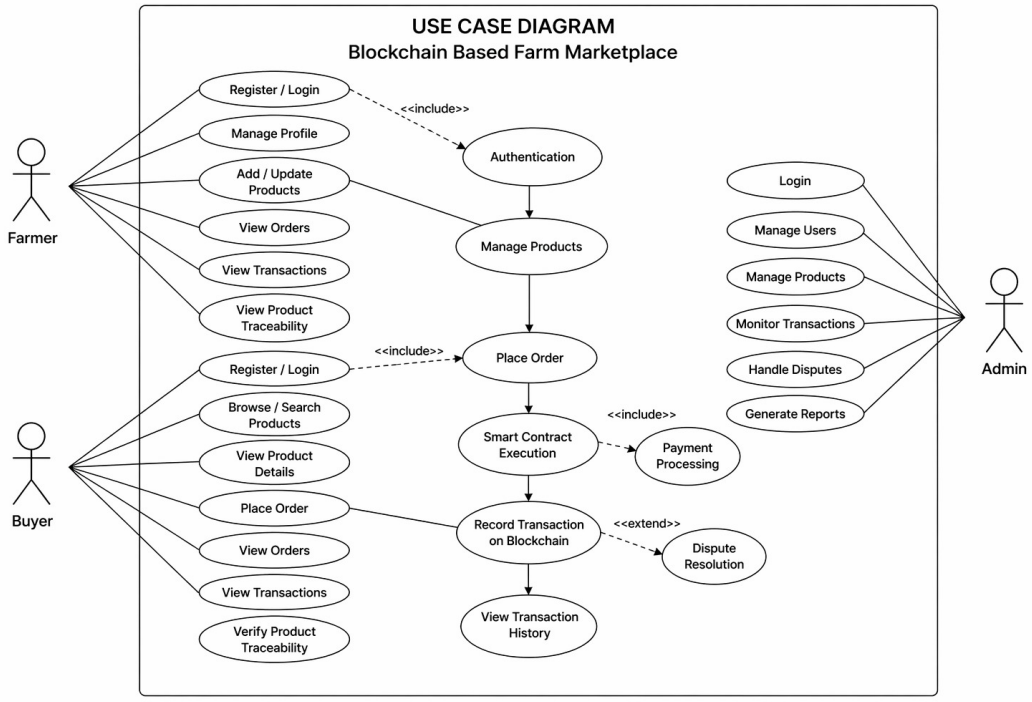


Figure 4.2: Use Case Diagram

4.3 Data Flow Design

When a buyer places an order, the request is submitted through the application to the Node.js backend. The backend verifies the request and the buyer's JWT, then triggers the corresponding smart contract function via Ethers.js through the relayer service. The smart contract locks the buyer's payment in escrow and records the order on the blockchain. The order status is updated in the MongoDB database and a notification is dispatched to the farmer. Once the buyer confirms delivery, the contract releases the escrowed payment to the farmer; if the order is cancelled, the funds are returned to the buyer. Traceability data linked to the product listing is retrieved from the blockchain upon buyer request and displayed on the product detail page.

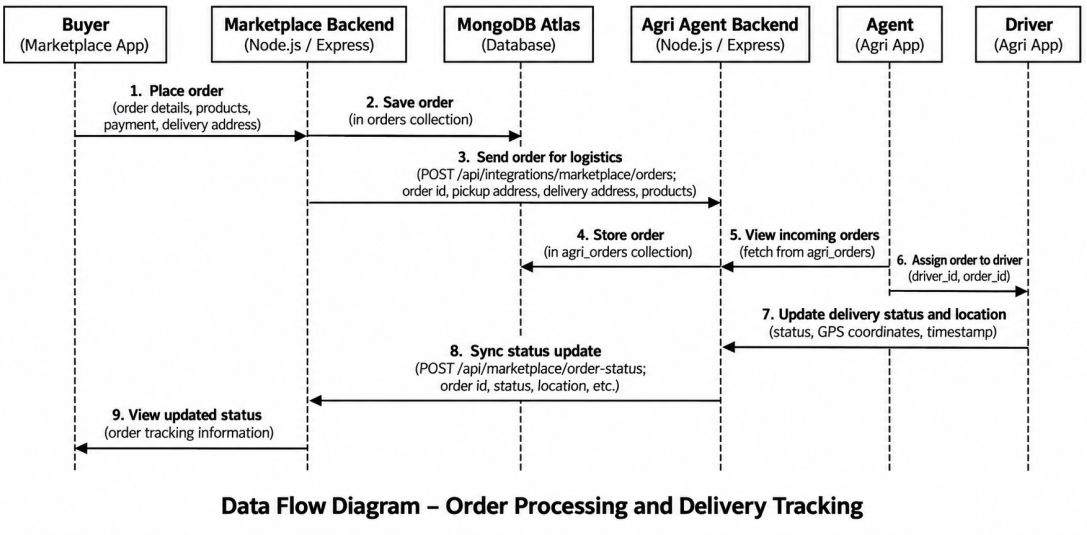


Figure 4.3: Data Flow Diagram

4.4 Database Design

The off-chain MongoDB database (accessed through Mongoose) stores user profiles (name, role, contact, hashed credentials), product listings (crop type, quantity, price, description, image URL, farmer ID), order records (buyer ID, product ID, quantity, status, timestamp), shopping cart records, and buyer reviews and ratings. On-chain data stored on the blockchain include transaction records (transaction ID, parties involved, amount, timestamp), smart contract state variables (product registry, order status, escrow payment status), and traceability logs linked to each product.

Implementation

5.1 Smart Contract Development

The smart contracts for the blockchain are written in Solidity which is deployed on an Ethereum compatible blockchain. The primary smart contract (FarmMarketplace.sol) is used to define the product listing structure, buyer-seller transaction logic, and payment escrow functionality. The Key functions are listProduct(), purchaseProduct(), confirmDelivery(), cancelOrder(), and getProductHistory(). Buyer payments are locked in escrow at purchase and released to the farmer only after delivery is confirmed. Hardhat is used as the development and testing framework, and contracts are deployed to a local Hardhat node during development.

5.2 Frontend Development

The frontend is built using Expo (React SDK 58) together with Expo Router, React Native Web and React Native Paper. The frontend runs on Android, iOS and the web from a codebase. There are three dashboards, for farmers, for buyers and for administrators. The frontend talks to the backend REST API. The users do not need a browser wallet. The frontend includes components, for product listing forms, marketplace browsing, shopping cart order tracking, reviews, image capture and upload and traceability detail pages.

5.3 Backend Development

The backend is built with REST API using Node.js, Express.js and TypeScript. The backend REST API handles user registration and authentication with JWT-based session management and role-based access control. The backend REST API also performs product CRUD operations for off-chain metadata manages orders supports a shopping cart processes reviews and provides endpoints, for government scheme guidance and AI guidance. The backend REST API interacts with blockchain through Ethers.js. Password hashing is implemented using bcrypt. Product images are uploaded to Cloudinary. The backend communicates with MongoDB (through Mongoose) for off-chain storage and with the Ethereum node for on-chain operations.

5.4 Blockchain Integration

Ethers.js 6 is configured in the backend to connect to a local Hardhat blockchain node during development. The contract ABI generated after Solidity compilation is imported into the backend to interact with the deployed contract. A backend relayer service derives a deterministic wallet address for each user and signs and submits transactions on the user's behalf, removing the need for users to manage wallets or pay gas fees directly. This design improves usability for farmers with limited technical knowledge, but it introduces a trust dependency on the backend, since private keys are managed server-side.

5.5 Authentication and Role-Based Access

The login interface it offers Farmer, Buyer and Admin roles. The users can sign in using an email address and password and with choose to remember this session to recover a forgotten password register an account pick a language or change the theme. The role that is selected determines which dashboard and management functions appear after the user signs, in. The resulting screen is shown in Figure 7.1.

5.6 Farmer Dashboard

The farmer dashboard provides a compact operational overview. It displays the number of products listed, total orders, average rating, and reviews. Quick actions provide direct navigation to product creation, product viewing, order management, blockchain transactions, and government schemes (Figure 7.2).

5.7 Product Listing and Image Upload

Farmers can create a new listing by entering product and farm information, including the farm location. Product images can be captured using the device camera, selected from the gallery, or supplied through an image URL. The final listing action, "List Produce & Secure On-Chain", connects the marketplace workflow with the blockchain transaction design described earlier (Figure 7.3).

5.8 Order Management

Incoming orders are presented as individual order cards containing the order identifier, date, buyer, payment method, product quantity, unit price, and order total. Pending orders can be accepted or rejected, and accepted orders can progress to shipment. This interface supports

the order-placement and transaction-history requirements defined in the SRS (Figure 7.4).

5.9 Buyer Marketplace Interface

The buyer interface provides marketplace browsing with a search field and category filters. Product cards show the product image, organic/database status, farmer, location, stock, and price. Buyers can add products to the cart or proceed directly with the Buy Now action (Figure 7.5).

5.10 Administrative Dashboard

The administrator dashboard provides platform-level information including total revenue, commission, current-month values, and counts of farmers and buyers. It serves as the entry point for monitoring marketplace activity and accessing management functions (Figure 7.6).

5.11 Product Management

The Product Management interface allows administrators to search and filter product records by status. Each product card displays the farmer, available stock, verification state, price, and approval status. Administrative controls allow a listing to be blocked or deleted (Figure 7.7).

5.12 User Management

The User Management interface displays registered users and supports filtering by role and account state. User cards show name, email, joining date, and role. For non-admin accounts, administrative actions include suspension and deletion, supporting the account-management requirements defined in this report (Figure 7.8).

System Testing

6.1 Testing Objectives

The objective of system testing is to verify that all functional and non-functional requirements of the Blockchain Based Farm Marketplace are correctly implemented and that the system behaves as expected under various conditions. Testing validates the correctness of smart contract execution, the accuracy of product listing and order management features, the security of user authentication, and the integrity of blockchain-recorded data.

6.2 Types of Testing Conducted

6.2.1 Unit Testing

Individual components, including smart contract functions, backend API endpoints, and frontend components, are tested in isolation. Smart contract unit tests are written in TypeScript (`FarmMarketplace.test.ts`) using the Hardhat testing framework with Mocha and Chai. API endpoints are tested using Postman. Each unit test verifies correct outputs for both valid inputs and edge cases.

6.2.2 Integration Testing

Integration testing verifies the end-to-end flow across the frontend, backend, and blockchain layers. Key scenarios include farmer product listing through the application triggering correct blockchain state changes, buyer order placement locking payment in escrow, delivery confirmation releasing funds to the farmer, and transaction history retrieval displaying accurate records on the dashboard.

6.2.3 User Acceptance Testing

User acceptance testing is conducted with representative farmer and buyer users to evaluate the usability and accessibility of the platform. Participants perform tasks such as registering an account, listing a product, placing an order, and checking transaction history. Feedback is used to refine the user interface and improve workflow clarity.

6.3 Test Cases

Table 6.1: Sample Test Cases

TC No.	Test Case	Input	Expected Output	Result
TC01	User Registration	Valid user details	Account created, role assigned	Pass
TC02	Login Authentication	Valid credentials	User logged in, JWT issued	Pass
TC03	Product Listing	Crop details, price, quantity	Product saved and recorded on blockchain	Pass
TC04	Order Placement	Product ID, buyer account	Payment locked in escrow, order recorded	Pass
TC05	Delivery Confirmation	Order ID	Escrowed funds released to farmer	Pass
TC06	Order Cancellation	Order ID	Escrowed funds returned to buyer	Pass
TC07	Product Traceability	Product ID	Origin and history displayed from blockchain	Pass
TC08	Invalid Login	Wrong password	Access denied, error message shown	Pass
TC09	Admin User Management	Admin credentials	User list displayed, management options available	Pass

Results and Discussion

This chapter presents the outputs obtained from the implemented system. Each section shows the corresponding application screen and discusses the result observed. The values displayed in the screens are snapshot data from the running application and should be treated as demonstration-state results rather than benchmark measurements.

7.1 Smart Contract Execution Results

Upon order placement, the backend relayer invokes the smart contract's `purchaseProduct()` function, locking the buyer's payment in escrow. The transaction is signed by the relayer, broadcast to the blockchain, and confirmed after mining. The resulting transaction hash, timestamp, buyer address, seller address, product ID, and amount are recorded immutably on the blockchain ledger and displayed to both parties. When the buyer confirms delivery, `confirmDelivery()` releases the escrowed funds to the farmer.

7.2 Traceability Verification

Buyers can access the traceability view for any product, which queries the blockchain for all recorded events linked to that product's ID. It displays the listing date, farmer identity (blockchain address), and transaction records, providing verifiable proof of origin without additional hardware.

7.3 Login Page Output

The login page allows users to sign in as a Farmer, Buyer, or Admin using their email and password. After successful authentication, the user is taken to the dashboard corresponding to the selected role.

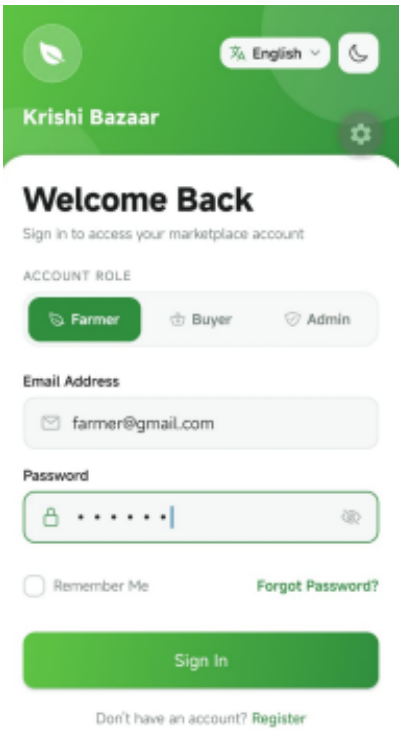


Figure 7.1: Login Page with Role Selection for Farmer, Buyer, and Admin

7.4 Farmer Dashboard Output

The farmer dashboard reports two listed products and six total orders in the captured state, along with the average rating and review count. Quick action buttons give direct access to product creation, order management, blockchain transactions, and government schemes.

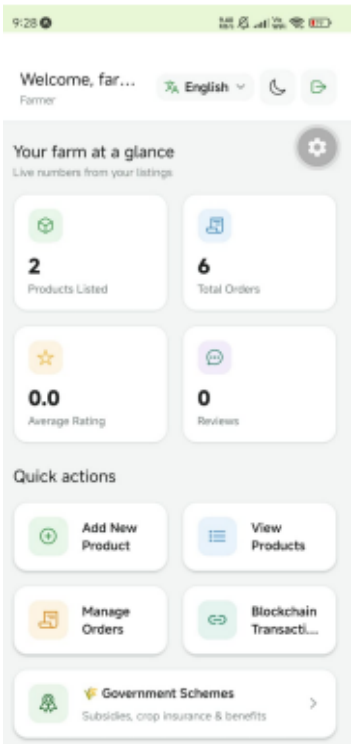


Figure 7.2: Farmer Dashboard with Product, Order, Rating, and Review Summaries

7.5 Product Listing Output

The Add New Product page lets the farmer enter product details and the farm location, and attach an image using the camera, gallery, or an image URL. On submission through the "List Produce & Secure On-Chain" action, the listing is stored off-chain in MongoDB and recorded on the blockchain through the smart contract via the backend relayer.

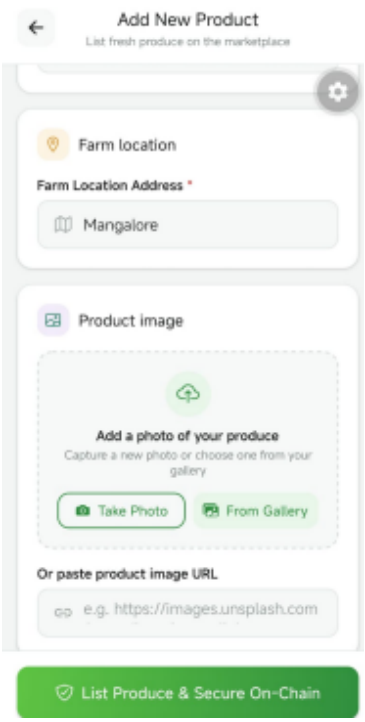


Figure 7.3: Add New Product Page with Farm Location and Image Capture/Upload Options

7.6 Incoming Orders Output

The Incoming Orders page shows each order as a card with the buyer, payment method, quantity, and total. The captured state contains pending potato orders with totals of Rs. 100 and Rs. 50, and another order totalling Rs. 250. The farmer can accept or reject pending orders and move accepted orders to shipment.

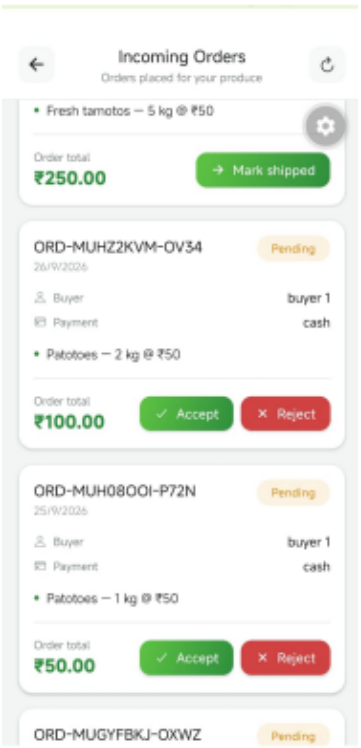


Figure 7.4: Incoming Orders Page with Accept/Reject and Shipment Actions

7.7 Buyer Marketplace Output

The buyer marketplace displays active listings with search and category filters. The captured state shows a "Fresh tamotos" listing at Rs. 50 per kilogram with 45 kg stock in Mangalore. Buyers can add products to the cart or use the Buy Now action, and can view reviews and blockchain traceability data for each product.

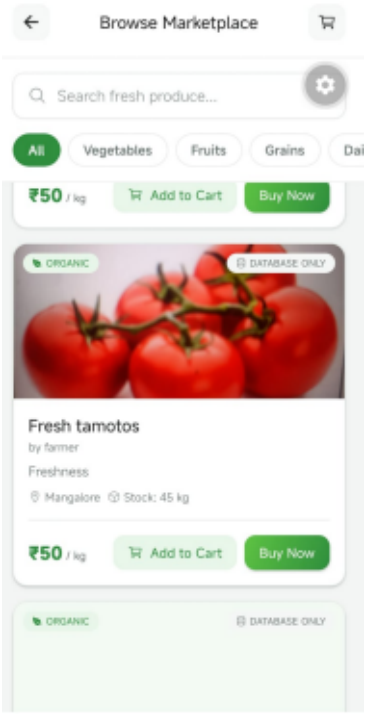


Figure 7.5: Buyer Marketplace with Search, Category Filters, Cart, and Buy Now Actions

7.8 Admin Dashboard Output

The admin dashboard reports total revenue of Rs. 380 and a commission of Rs. 8, with three farmers and four buyers in the overview. It acts as the entry point for monitoring marketplace activity.

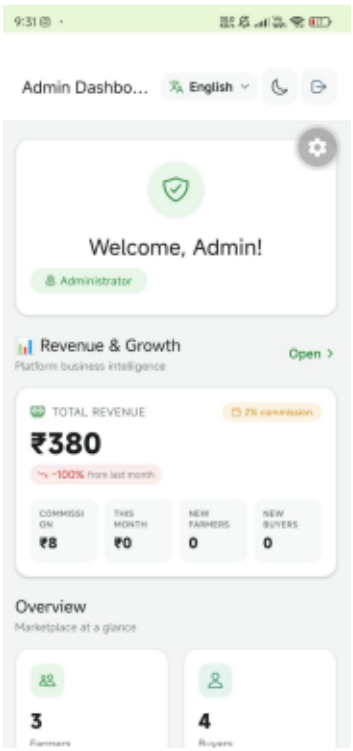


Figure 7.6: Admin Dashboard with Revenue, Commission, Farmer, and Buyer Summaries

7.9 Admin Product Management Output

The Product Management page lists approved listings along with farmer, stock, verification state, and price. The administrator can search, filter by status, and block or delete a listing.

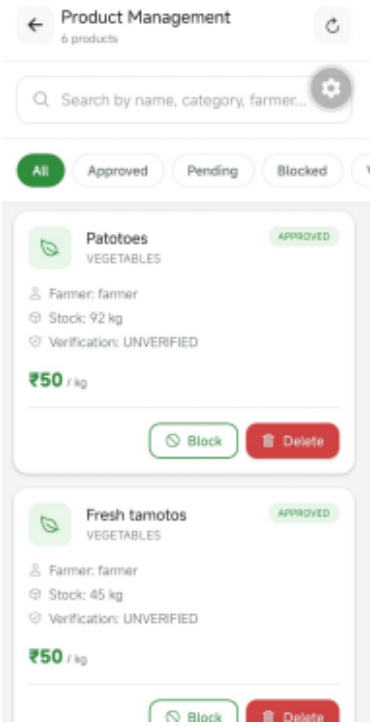


Figure 7.7: Admin Product Management with Status, Stock, Verification, Block, and Delete Controls

7.10 Admin User Management Output

The User Management page shows role-labelled accounts with name, email, and joining date, and supports filtering by role and account state. The administrator can suspend or delete non-admin accounts.

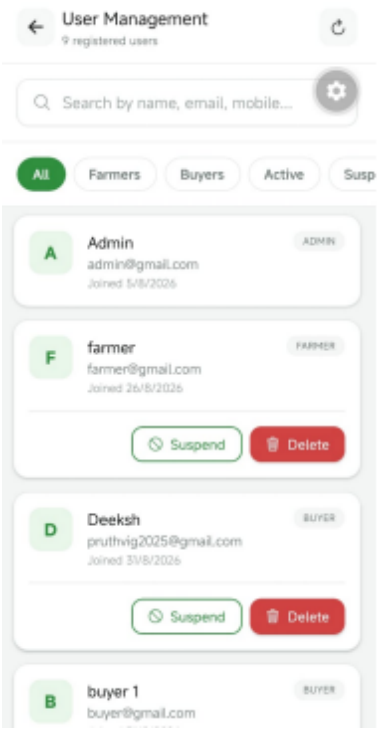


Figure 7.8: Admin User Management with Role Filters and Suspend/Delete Controls

7.11 Discussion

Together, these outputs show that the platform provides separate operational and administrative views for farmers, buyers, and administrators rather than relying on a single generic interface, and that the escrow-based order flow and traceability records are handled through the smart contract and relayer. The results indicate that the system meets its usability and workflow objectives, while performance at large scale and on public networks remains subject to further evaluation.

Conclusion and Future Work

8.1 Conclusion

The Blockchain Based Farm Marketplace for Transparent Agricultural Trade is a platform that was designed and built to connect farmers directly with buyers removing old middlemen. It uses Ethereum contracts to automate trade agreements enforce payment rules and record all transactions immutably on the blockchain, which provides farmers direct market access and fair pricing while buyers receive product information and verifiable traceability data. It uses Ethereum contracts to make trade agreements automatic to enforce payment rules and to write every transaction forever on the blockchain. The Blockchain Based Farm Marketplace gives farmers access to markets and fair prices. Buyers get product details and can verify where the product came from. The current version signs transactions through a backend relayer, which makes the user experience easier. Adds some central control.

The software requirements for the Blockchain Based Farm Marketplace have been comprehensively defined. They include user registration and authentication farmer product listing buyer marketplace browsing, smart contract-based order placement blockchain transaction recording and admin management. Non-functional requirements such, as security, reliability, scalability and usability are addressed through the chosen technology stack consisting of Expo (React Native) Node.js, TypeScript, MongoDB, Solidity, Ethers.js and the Ethereum blockchain. The system design is presented through an architecture diagram a use case diagram and a data flow diagram and Blockchain Based Farm Marketplace demonstrates that the Blockchain Based Farm Marketplace can be built as planned.

8.2 Future Work

The proposed system can be extended in several directions. The cross-platform application can be published to the Google Play Store and Apple App Store to reach farmers in rural areas, and regional language support can be added to improve accessibility for farmers across diverse linguistic backgrounds in India. IoT sensors can be incorporated for real-time monitoring of product conditions such as temperature, humidity, and storage environment, with the readings recorded on the blockchain, while the prototype AI guidance module can be extended with machine learning models for demand forecasting, price recommendation, and fraud detection. The platform can also be connected with national agricultural data registries

and e-NAM portals to extend its reach and regulatory compliance. On the blockchain side, Layer-2 scaling solutions such as Polygon can be adopted to reduce gas fees and improve transaction throughput for large-scale deployments, and the backend relayer can be replaced with user-controlled wallets to reduce the trust and centralization dependency.

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